ALEIX CASAS BARCO

Full Stack Software Engineer

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SUMMARY

Full Stack Software Engineer with 3 years in the product team of a SaaS urban analytics platform, where I have owned four projects end to end and mentored two junior developers. I built the platform's AI report generation from scratch, converted the application into an embeddable widget now running in production, and work daily on Django REST APIs over PostgreSQL holding tens of millions of records. MSc in Software Development and Architecture (La Salle, Universitat Ramon Llull, 2026).

SKILLS

- **Languages:** Python, TypeScript, JavaScript, SQL
- **Backend:** Django, Django REST Framework, Celery, Node.js, REST API design, asynchronous processing
- **Frontend:** Angular (signals, RxJS, testing), React, HTML, CSS
- **AI:** LLM integration in production (Google Gemini), context layer design, automated report generation
- **Databases:** PostgreSQL, Redis, BigQuery, data modeling, query optimization, indexing, partitioning
- **Cloud and Infrastructure:** Google Cloud Platform (GKE, Cloud SQL, BigQuery), Kubernetes (HPA, KEDA), Terraform, Docker, GitHub Actions, CI/CD
- **Observability and Testing:** Prometheus, Grafana, load testing, unit, integration and end-to-end testing
- **Architecture:** Microservices, design patterns, distributed systems, event-driven architecture, scalable system design
- **Ways of working:** Agile, Scrum, Git, Jira, code review, mentoring

EXPERIENCE

Full Stack Software Engineer — Idencity, Barcelona | June 2023 - Present

Product team of 3 engineers building IDDIGO, a SaaS urban analytics platform with over 1000 city and territory performance indicators.

- Owned 4 projects end to end, from technical design and implementation through to launch:
  - **AI report generation.** Designed and built from scratch the layer that gathers and structures the platform's full data context and calls Google Gemini to produce analysis reports. What users previously had to write by hand is now generated in seconds.
  - **Embeddable widget.** Converted the Angular application into a component that can be embedded in third-party websites, which required a new authentication system and reworking parts of the frontend. Live in production.
  - **Customizable territorial profile.** User-configurable product feature.
  - **Migration of the Angular application from imperative to reactive.** Moved the codebase to signal-based state and Angular's modern syntax, and built the testing system from zero — unit and end-to-end — on a project that had no tests, along with the infrastructure and conventions for the team to keep writing them.
- Designed and built **Django REST APIs over PostgreSQL holding tens of millions of records**, optimizing queries and indexes to cut the heaviest endpoints from several seconds to under one.
- Modeled complex hierarchical data structures (trees and graphs) in PostgreSQL with custom algorithms for creation, traversal and efficient manipulation.
- Implemented asynchronous processing with Celery for report generation and bulk data jobs, moving blocking work off the critical path.
- Built scalable Angular components with advanced pagination, infinite scroll and Highcharts data visualization.
- Designed the standardized data export system (CSV, Excel, PDF) with traceability and architectural consistency, reused across the platform.
- Mentored 2 junior developers through code reviews, pairing and technical growth plans.

MASTER'S THESIS PROJECT

Vista Flota — Real-time IoT fleet management platform on Google Cloud | La Salle URL, 2026

Vista Flota lets fleet operators follow their trucks in real time. An IoT device on each vehicle publishes telemetry over MQTT — position, speed and fuel — which the platform ingests, deduplicates and turns into live fleet state, trips and alerts on an operator dashboard, while archiving the full history for route reconstruction and analysis.

Team project across three areas (Application, Infrastructure, IoT). Owner of the Application area.

- Built the event processor and data layer of a real-time ingestion system and took it **from 80 to 5,000 events per second at peak and 3,200 sustained with 0% message loss**, against a requirement of 1,000 events per second.
- Diagnosed and removed bottlenecks through a measure, diagnose, fix and re-validate cycle instrumented with Prometheus and Grafana: moved deduplication out of PostgreSQL into Redis to clear the critical path, added PgBouncer multiplexing 200 client connections over 20 real ones, implemented watermark-based backpressure to eliminate out-of-memory crashes, and switched MQTT to QoS 1 to stop message loss on pod restarts.
- Replaced KEDA autoscaling with Kubernetes HPA after profiling proved the processor pods were CPU-bound.
- Designed polyglot persistence: Redis for hot fleet state under 1 ms, Cloud SQL for relational data, and date-partitioned BigQuery for historical telemetry.
- Built the Angular operator dashboard and the backend REST API.
- **Stack:** TypeScript, Angular, Kubernetes (GKE), Terraform, MQTT/EMQX, Cloud SQL, BigQuery, Redis, Prometheus, Grafana, GitHub Actions.

EDUCATION

MSc in Software Development and Architecture — La Salle, Universitat Ramon Llull | Oct 2025 - Jul 2026

Software architecture, microservices, scalable system design, Kubernetes, NoSQL, testing, agile methodologies, AWS.

BSc in Computer Engineering — Universitat Autonoma de Barcelona | 2020 - 2024

LANGUAGES

Catalan (native) | Spanish (native) | English (B2, professional working proficiency)